

DJ NETWORK'S RENTAL DATABASE ENGINE

MIS Functional Requirement & Implementation Roadmap

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1. Purpose of This Document

This document defines the functional requirements, database structure, business logic, workflow, controls, reporting requirements, and implementation roadmap for a simple but scalable rental tracking engine for DJ Network's. The objective is to create one reliable source of truth for equipment rentals while keeping daily staff operations fast and simple.

The system should initially be implemented using Google Sheets/Google Forms/Google Drive and may later be upgraded to an AppSheet or custom database application without changing the underlying business logic.

2. Core Design Principle

- Google Sheet = official database / source of truth.
- Google Form = simple mobile data-entry interface for staff.
- Google Drive = storage for rental, equipment, ID-proof and condition photographs/documents.
- WhatsApp = quick operational communication and photo trail; it must NOT be the master database.
- Every rental transaction receives a unique Rental ID.
- Every physical equipment asset receives a permanent Equipment ID.
- One rental order may contain multiple equipment lines.
- Order person and delivery/collection person must be tracked separately.
- Dispatch and return condition must both be recorded.
- The system must retain historical records; old rental records should never be overwritten or deleted merely because equipment is returned.

3. High-Level Workflow

MASTER DATA → RENTAL ORDER → EQUIPMENT ALLOCATION → DISPATCH → DELIVERY/PICKUP → ACTIVE RENTAL → RETURN → CONDITION CHECK → DAMAGE/SHORTAGE → PAYMENT CLOSURE → EQUIPMENT AVAILABLE → REPORTING/ANALYTICS

Recommended daily flow:

1. Customer/order is received.
2. Customer is searched or created in Customer Master.
3. Rental Order is created and assigned a Rental ID.
4. Required equipment is selected from Equipment Master.
5. Availability is checked against existing active/reserved rentals.
6. Delivery method and delivery person are entered, if applicable.
7. Dispatch photographs and documents are captured.

8. Equipment status changes from Available/Reserved to Rented Out.
9. WhatsApp dispatch message is posted using the standard format.
10. On return, the same Rental ID is used to record return date, condition, damage and missing items.
11. Equipment status changes back to Available, Maintenance, Damaged or Lost depending on inspection.
12. Payment/security-deposit status is closed.
13. Dashboard and equipment rental history update automatically.

4. Database Structure

4.1 EQUIPMENT MASTER

Purpose: Permanent master list of all rentable assets and inventory.

Field	Required?	Example	Logic
Equipment ID	Yes	EQ-0001	Permanent unique ID; never reused.
Equipment Name	Yes	Pioneer CDJ-3000	Standardized name only.
Category	Yes	DJ	Dropdown.
Brand	Yes	Pioneer DJ	Master value.
Model	Yes	CDJ-3000	Standard model.
Serial Number	Recommended	ABC12345	Unique where applicable.
Quantity	Yes	2	For identical units managed as a quantity pool.
Rental Rate	Yes	8000	Default rate; actual order rate can differ.
Current Status	Yes	Available	Calculated/controlled status.
Notes	No	Flight case included	Free text.

Sample entries:

Equipment ID	Equipment Name	Category	Brand	Model	Qty	Status
EQ-0001	Pioneer CDJ-3000	DJ	Pioneer DJ	CDJ-3000	2	Available
EQ-0002	Pioneer DJM-A9	DJ	Pioneer DJ	DJM-A9	1	Available
EQ-0003	Traktor S8	DJ	Native Instruments	Traktor S8	1	Available
EQ-0004	Yamaha Montage M6	Keyboard	Yamaha	Montage M6	1	Available

4.2 CUSTOMER MASTER

Purpose: Avoid repeatedly entering the same customer/company information for every rental.

Field	Required?	Example	Logic
Customer ID	Yes	CUS-0025	Permanent unique customer ID.
Customer/Company Name	Yes	ABC Events Pvt. Ltd.	Standard legal/business name where available.
Contact Person	Yes	Rahul Mehta	Primary operational contact.
Mobile	Yes	98XXXXXXX	Primary contact number.
Address	Recommended	Ahmedabad	Billing/delivery reference.
ID Proof Type	Conditional	Aadhaar / PAN / Driving	Capture only when required

		Licence	by company policy.
ID Number	Conditional	XXXX	Restrict access; do not expose unnecessarily.
GSTIN	Optional	24XXXXXXXXXXXXZ5	For business customers where applicable.
Notes	No	Regular corporate client	Free text.

4.3 RENTAL ORDER

Purpose: One record for the commercial transaction. One Rental ID can contain multiple equipment items.

Field	Example	Logic
Rental ID	RNT-2026-0001	Auto-generated; unique.
Date Out	02-09-2026	Actual dispatch/pickup date.
Date In	04-09-2026	Actual return date; blank while active.
Customer ID	CUS-0025	Linked to Customer Master.
Customer/Company	ABC Events Pvt. Ltd.	Auto-populated from Customer Master.
Contact Person	Rahul Mehta	Auto-populated or editable.
Mobile	98XXXXXXXX	Auto-populated.
Address	Ahmedabad	Delivery/reference address.
ID Proof Type	Aadhaar	Only if required.
ID Number	XXXX	Access controlled.
Order By	Niraj	Person who accepted/created the booking.
Delivery Mode	Porter	Dropdown.
Delivery Person	Ramesh	Separate from Order By.
Delivery Mobile	98XXXXXXXX	Optional/required when operationally useful.
Rental Amount	15000	Actual agreed rental.
Security Deposit	10000	Track separately from rental revenue.
Payment Status	Paid	Paid/Partial/Pending.
Equipment Condition Out	Good	Overall summary; item-level condition may also be stored.
Equipment Condition In	Good	Overall summary.
Damage	No	No/Yes.
Notes	Corporate Event	Special instructions.
Photo/Document Link	Drive URL	Link to folder/photos.

4.4 RENTAL ITEMS / EQUIPMENT ALLOCATION

This is required for correct equipment-level history. Do not force multiple products into one cell. Each equipment line should be a separate row linked to the same Rental ID.

Rental ID	Equipment ID	Equipment Name	Qty	Dispatch Condition	Return Condition	Damage	Damage Charge
RNT-2026-0001	EQ-0001	Pioneer CDJ-3000	2	Good	Good	No	0
RNT-2026-0001	EQ-0002	Pioneer DJM-A9	1	Good	Damaged	Yes	5000

5. Equipment Availability Logic

Before confirming a rental, the engine must calculate whether the requested quantity is available for the requested date range.

- Available quantity = Master quantity – quantity already allocated to overlapping active/reserved rentals.
- The system must check Date Out and Expected/Actual Date In overlap.
- A rental must not reduce inventory below zero.
- If insufficient stock exists, show an alert: NOT AVAILABLE / PARTIAL AVAILABILITY.

- Maintenance, Damaged and Lost quantities must not be counted as available.
- Returned equipment becomes Available only after return inspection is completed.
- A Reserved order blocks availability even before physical dispatch.

Example: EQ-0001 has Qty 2. If one unit is already reserved for 10–12 Sep and a new customer requests 2 units for 11 Sep, the system must flag insufficient availability.

6. Rental Status Logic

Status	Meaning	Trigger
Draft	Order being prepared	Created but not confirmed
Reserved	Equipment blocked for future rental	Booking confirmed
Rented Out	Equipment physically dispatched	Dispatch completed
Returned - Inspection Pending	Physical return received	Return recorded
Available	Ready for next rental	Inspection passed
Maintenance	Not rentable	Maintenance flag
Damaged	Not rentable until resolved	Damage recorded
Lost	Unavailable	Loss recorded
Cancelled	Order cancelled	Cancelled before dispatch

7. Delivery & Collection Tracking

Delivery/collection must be separate from the order creator. This prevents confusion when one employee takes the booking and another person handles transport.

Type	Method	Person	Mobile	Date	Order ID	Cost	Proof
Dispatch	Porter	Ramesh	98XXXXXXXX	02-09-2026	RNT-2026-0001	450	Receipt/Photo Link
Return	Self Delivery	Mahesh	98XXXXXXXX	04-09-2026	RNT-2026-0001	0	Photo Link

8. Photo & Document Management

Recommended Google Drive folder structure:

- DJ NETWORK'S RENTAL
- 2026
- RNT-2026-0001
- 01 - ID Proof
- 02 - Dispatch Photos
- 03 - Delivery Proof
- 04 - Return Photos
- 05 - Damage / Repair Evidence
- 06 - Bills / Receipts

The Sheet should store links, not large images. WhatsApp may contain the quick photo trail, while Drive stores the organized evidence.

9. WhatsApp Operating Standard

WhatsApp group: DJ NETWORK'S – RENTAL CONTROL

Dispatch template:

● RENTAL OUT

Rental ID: RNT-2026-0001

Customer: ABC Events

Mobile: 98XXXXXXX

Equipment: CDJ-3000 × 2; DJM-A9 × 1

Date Out: 02-09-2026

Return: 04-09-2026

Delivery: Porter

Delivery By: Ramesh

Order By: Niraj

Amount: ₹15,000

Deposit: ₹10,000

Condition: Good

Photos: Attached

Return template:

● RENTAL RETURN

Rental ID: RNT-2026-0001

Equipment: CDJ-3000 × 2; DJM-A9 × 1

Returned: 04-09-2026

Condition: Good

Damage: No

Received By: DJ Network's

Photos: Attached

The Rental ID is mandatory in every WhatsApp operational message so staff can search the corresponding database record instantly.

10. Data Validation & Controls

- Rental ID must be unique.
- Equipment ID must be unique and permanent.
- Customer ID must be unique.
- Equipment Name should be selected from a controlled master list, not free-typed.
- Delivery Mode, Payment Status, Damage and Equipment Status should use dropdowns.
- Date In cannot be earlier than Date Out.
- Return cannot be completed without identifying the Rental ID.
- Damage = Yes should require a damage note and preferably a photo link.
- Payment Status = Partial/Pending should be visible on dashboard.
- Security Deposit should never be included in rental revenue unless business policy explicitly treats it as income.
- Historical transactions should be append-only or protected from accidental edits.
- Sensitive ID-proof information should be access restricted and retained only according to the company's document-retention policy.

11. Automatic Reports / Dashboard

- Total equipment quantity.
- Available quantity.
- Currently rented quantity.
- Reserved quantity.
- Maintenance/damaged/lost quantity.
- Total rental orders.
- Current month rental orders.

- Current month rental revenue.
- Pending payments.
- Overdue returns.
- Equipment-wise number of rentals.
- Equipment-wise total rental days.
- Equipment-wise rental revenue.
- Customer-wise rental count.
- Customer-wise total rental value.
- Damage incidents by equipment.
- Last rental date for each equipment.
- Top rented equipment for the year.
- Equipment with low/no utilization.

12. Equipment Utilization Logic

The engine must preserve equipment-level history so management can answer: 'How many times did this product go on rental this year?'

Equipment ID	Year	Rental Count	Total Rental Days
EQ-0001	2026	43	96
EQ-0002	2026	37	82
EQ-0003	2026	21	44

Rental Count should count completed/active rental allocations for that equipment. Total Rental Days should be calculated from the agreed rental period, with a clearly documented convention for same-day rentals.

13. Search & History Requirements

The MIS/database should provide fast search by:

- Rental ID
- Customer / Company
- Mobile number
- Equipment ID
- Equipment Name
- Serial Number
- Date range
- Order By
- Delivery Person
- Payment Status
- Rental Status

Example: Searching EQ-0001 should show its complete rental history, including every customer, date out, date in, quantity, condition, damage and rental amount associated with that asset.

14. Google Sheets Implementation

Minimum viable workbook tabs:

- RENTAL RECORD
- EQUIPMENT LIST
- CUSTOMER MASTER
- RENTAL ITEMS
- LOGISTICS

- DASHBOARD
- LISTS / SETTINGS (hidden or protected)

For the simplest staff interface, create two Google Forms: NEW RENTAL / DISPATCH and RENTAL RETURN. Form responses should feed protected raw-data sheets; formulas and dashboards should operate on structured tables.

15. Recommended Automation Rules

Automation	Required Logic
Rental ID	Generate automatically as RNT-YYYY-####.
Customer ID	Generate automatically as CUS-#### when a new customer is created.
Equipment ID	Assign once in Equipment Master; never change.
Availability	Calculate from overlapping reservations/rentals.
Status	Change automatically based on reservation, dispatch and return events.
Overdue	If today's date > expected return date and Date In is blank, mark OVERDUE.
Damage	If Damage = Yes, flag equipment for inspection and exclude it from Available quantity.
Payment	Pending/Partial rows should appear in dashboard alerts.
Utilization	Count rental allocations and rental days by Equipment ID and year.
Audit	Keep Created By, Created Date, Last Updated By and Last Updated Date where technically feasible.

16. Roadmap for MIS Development

Phase 1 – Database Foundation

Create master tables, unique IDs, dropdown lists, protected columns and standard naming conventions.

Output: Clean Google Sheets database.

Phase 2 – Rental Workflow

Create New Rental/Dispatch and Return forms; implement order, equipment allocation, customer and logistics flow.

Output: Staff can execute the full rental lifecycle.

Phase 3 – Validation & Availability

Add date-overlap availability checks, status controls, overdue flags, duplicate checks and mandatory fields.

Output: System prevents common operational errors.

Phase 4 – Documents & WhatsApp

Create Drive folder logic, document links and standardized WhatsApp templates using Rental ID.

Output: Searchable operational evidence.

Phase 5 – Dashboard & Analytics

Build KPI dashboard, equipment utilization, customer history, revenue and exception reports.

Output: Management visibility.

Phase 6 – Testing & Handover

Test real scenarios, reconcile inventory, lock master formulas, train staff and document SOP.

Output: Production-ready basic rental engine.

Phase 7 – Future Upgrade

If volume increases, migrate the same logic to AppSheet/custom database/ERP while retaining IDs and historical data.

Output: Scalable application.

17. Test Scenarios MIS Must Validate

1. One equipment item rented to one customer.
2. Multiple equipment items under one Rental ID.
3. Two identical units rented from the same equipment pool.
4. Overlapping booking where insufficient quantity exists.
5. Self pickup with no delivery person.
6. Porter/Rapido delivery with a separate delivery person.
7. Order created by one person and delivered by another.
8. Rental returned on time.
9. Rental returned late.
10. Equipment returned damaged.
11. Equipment returned with missing quantity.
12. Partial payment.
13. Pending payment.
14. Security deposit recorded and closed separately.
15. Rental cancelled before dispatch.
16. Equipment sent to maintenance after return.
17. Equipment history searched by Equipment ID.
18. Customer history searched by Customer ID/mobile.
19. Yearly rental count generated correctly.

18. Sample End-to-End Transaction

Scenario: ABC Events rents 2 × Pioneer CDJ-3000 and 1 × Pioneer DJM-A9 for a corporate event.

Step	Sample Data
Customer	CUS-0025 / ABC Events / Rahul / 98XXXXXXXXX
Rental ID	RNT-2026-0001
Period	02-09-2026 to 04-09-2026
Equipment	EQ-0001 × 2; EQ-0002 × 1
Order By	Niraj
Delivery	Porter / Ramesh
Rental Amount	₹15,000

Security Deposit	₹10,000
Dispatch	Good condition + photos
Return	Good condition + photos
Damage	No
Final Status	Available

19. MIS Deliverables Checklist

- ☐ Google Sheet database with protected master structure.
- ☐ Customer Master.
- ☐ Equipment Master.
- ☐ Rental Record.
- ☐ Rental Items / Equipment Allocation.
- ☐ Logistics tracking.
- ☐ Dropdown/list configuration.
- ☐ Automatic unique IDs.
- ☐ Availability and overlap logic.
- ☐ Return/damage logic.
- ☐ Overdue logic.
- ☐ Google Forms for dispatch and return.
- ☐ Google Drive document/photo structure.
- ☐ Dashboard.
- ☐ Equipment rental frequency report.
- ☐ Customer history report.
- ☐ Payment pending report.
- ☐ Basic user SOP.

20. Final Implementation Principle

The system must remain simple for the staff member entering a rental but structured enough for management reporting. The staff member should never have to understand database relationships. They should only select the customer, equipment, dates, delivery method, responsible people, amount and condition. The engine should handle IDs, availability, status, history, calculations and dashboard reporting automatically.

Priority should be given to data accuracy, standardized Equipment IDs, Rental IDs, historical traceability and simple mobile entry. Do not over-engineer Version 1. Build the core rental lifecycle correctly first, then add automation and analytics.